Anh Ho

678-644-5003 | hoanh@stanford.edu | linkedin.com/in/anhho123 | github.com/onhole

Education

Stanford University — Stanford, CA
B.S. in Electrical Engineering, QuestBridge and Vida Valiente Scholar, GPA: 3.8 — *Jun 2028*

Georgia Institute of Technology — Atlanta, GA
Dual Enrollment in Mathematics, GPA: 4.0 — *Aug 2024 - May 2025*

Experience

R&D Digital Signal Processing Intern — Jun 2026 – Sep 2026
Keysight Technologies — *Colorado Springs, CO*

- Achieved live signal separation on a single twisted-pair, full-duplex transmission line at 1 GHz bandwidth without utilizing directional couplers.
- Recovered 100% live Ethernet packets while only discarding 0.026% of a 2ms sample.
- Scaled the separation ability 11.25× from 66.67 to 750 MBd symbol rate while maintaining 100% accuracy against IEEE 802.3bp's Transmitter Distortion test.
- Optimized FIR filter implementation from $O(N^2)$ to $O(N \log N)$ time complexity utilizing FFT.
- Designed a chip-independent, convenient, and IEEE-universal customer solution.

PCB Designer & Financial Officer — Feb 2026 – Present
Stanford Sensor Mesh Network (SMESH) — *Stanford, CA*

- Develop sensor nodes to measure environmental conditions for informed prescribed burning.
- Designed a breakout PCB for the Heltec V3 and connected sensors and power interfaces to significantly decrease bring-up cost and time. [GitHub Repository]

Embedded Software Engineer — Jan 2026 – Present
Stanford Solar Car Project — *Stanford, CA*

- Refactored C code to Rust, improving code reliability.
- Implemented temperature-controlled PWM fans to reduce car's power consumption.

Projects

Variable Power Supply | *Analog circuit design, oscilloscope*

- Designed and built an adjustable DC bench supply.
- Characterized regulation and ripple across loads (10k ohms–100 ohms), achieving 17.3–17.9 V output with max ripple as low as 1.9 mV.
- Validated operation up to ∼173 mA load current (3.0 W delivered at 17.3 V into 100 ohms) and documented results in a lab report (schematics + measurements).

Bare-Metal Bring-Up | *STM32, Cortex-M4, C/C++, PS Script*

- Authored custom startup code and linker script, implementing fully bare-metal firmware with direct register manipulation, deterministic timing, and cycle-level control.
- Wrote SysTick and TIM drivers for PWM generation and timing control without HAL/RTOS.
- Debugged at register level using GDB and OpenOCD.

Technical Skills

Languages: C, C++, Python, MATLAB, Rust
Hardware: STM32, ARM Cortex-M4, Heltec V3 (ESP32)
Tools: Altium Designer, KiCad, Keysight Infiniium, LTSpice, GDB, OpenOCD, Git/GitHub

Classes

Engineering: Circuits I (EE101A), Physics Electromagnetism (Physics 43), Physics Mechanics (Physics 41)
Computer Science: Programming Abstraction (CS106B), Harvard's Introduction to CS, Cybersecurity, and AI